Masar: Transformer-Based Pilgrim Tracking and Real-Time Safety System for Hajj Supervision

1st Faisal Almars (2240087)
Department of Computer Science and Artificial Intelligence
University of Jeddah
Jeddah, Saudi Arabia

2nd Abdulaziz Etaiwi (2140040)
Department of Computer Science and Artificial Intelligence
University of Jeddah
Jeddah, Saudi Arabia

3rd Omar Alshargawi (2240754)
Department of Computer Science and Artificial Intelligence
University of Jeddah
Jeddah, Saudi Arabia

4th Abdulrahman Alsumairi (2243821)
Department of Computer Science and Artificial Intelligence
University of Jeddah
Jeddah, Saudi Arabia

5th Safwan Alimam (2242740)
Department of Computer Science and Artificial Intelligence
University of Jeddah
Jeddah, Saudi Arabia

***Abstract*—Supervising pilgrims during Hajj is challenging because of crowd density, continuous group movement, and the difficulty of manually identifying separated or at-risk pilgrims. Traditional supervision methods such as roll calls, phone communication, and visual observation are limited in crowded environments and may delay response to abnormal movement. These delays can become serious when a pilgrim moves away from the group, stops unexpectedly, or follows a different path without the supervisor noticing immediately.**

This paper presents Masar, a real-time pilgrim tracking and safety system based on sequence-model anomaly detection. The system combines GPS tracking, a supervisor dashboard, backend services, and an artificial intelligence module that analyzes movement behavior. The current model comparison focuses on binary anomaly detection, where each movement sequence is classified as Normal or Abnormal.

Several sequence models were evaluated, including LSTM, RNN, and Transformer, in addition to a distance-threshold baseline. The Transformer model achieved strong anomaly detection performance with high recall and the highest ROC-AUC score among the evaluated methods. Experimental results on unseen routes demonstrate that sequence-based models can capture movement patterns more effectively than simple distance-based rules. The results show that Masar can provide supervisors with earlier awareness of risky movement, helping improve safety, organization, and response efficiency during large-scale pilgrim movement.

***Index Terms*—Hajj, pilgrim tracking, Transformer, anomaly detection, GPS tracking, time-series classification, real-time monitoring, crowd safety**

I. Introduction

Hajj is one of the largest annual gatherings in the world, where millions of pilgrims move between important locations such as Makkah, Mina, Arafat, and Muzdalifah. During these movements, campaign supervisors must ensure that pilgrims remain within their assigned groups and that separated or at-risk individuals are identified quickly. The scale of the event makes supervision difficult because movement is continuous, paths are crowded, and the same supervisor may be responsible for a large number of pilgrims.

In many campaigns, supervision still depends on manual techniques such as headcounts, phone calls, and messaging applications. These methods are simple, but they become unreliable in crowded and fast-moving environments. A pilgrim may drift away slowly, stop unexpectedly, follow a different path, or require help. Manual detection of these cases can be delayed, especially when one supervisor is responsible for many pilgrims. Even when supervisors are experienced, it is difficult to observe every individual continuously.

GPS tracking can improve awareness, but raw location points alone are not enough. A supervisor may see where a pilgrim is, but still needs to decide whether the movement is normal or abnormal. For example, a pilgrim may be slightly far but returning to the group, while another may be closer but moving away quickly. This means the system must analyze movement over time, not only a single location. A single point may look safe, while the full sequence may show a dangerous trend.

Masar addresses this issue by combining real-time tracking with sequence-based anomaly detection. The system analyzes recent movement readings and classifies pilgrim behavior as Normal or Abnormal. The goal is to support supervisors with automatic risk awareness and reduce dependence on manual monitoring. Instead of asking supervisors to manually interpret every GPS point, Masar highlights the cases that require attention.

This paper evaluates multiple sequence models and demonstrates that Transformer-based approaches provide strong performance for anomaly detection in pilgrim movement data. The main contribution of this work is the design and evaluation of a practical AI-supported tracking system that uses engineered movement features and sequence modeling to improve supervision during Hajj.

II. Background and Motivation

Large-scale crowd management requires fast decision-making and clear situational awareness. In Hajj campaigns, the supervisor must know whether each pilgrim is moving with the group, falling behind, taking a different route, or experiencing an emergency. The difficulty is not only in collecting location data but also in understanding what that data means.

A common approach to tracking is to use a fixed distance threshold. If a pilgrim moves more than a certain distance away from the supervisor, the system marks the pilgrim as out of range. This approach is easy to implement, but it has several weaknesses. First, a pilgrim may temporarily move outside the threshold because of crowd flow but then return to the group. Second, a pilgrim may still be inside the threshold but moving in the wrong direction. Third, distance alone does not represent speed, direction, or movement consistency.

Sequence modeling is suitable for this problem because movement is temporal. The meaning of a GPS point depends on previous points. For example, if the distance from the supervisor increases gradually over several readings, this may indicate separation. If the direction changes suddenly, the pilgrim may have taken a wrong path. If the time gap between readings is large, the system may need to treat the prediction with lower confidence. Therefore, the model should learn from windows of recent movement rather than isolated readings.

Masar is motivated by the need for a system that is simple enough for real supervisors to use but intelligent enough to detect risky movement patterns. The system does not replace the supervisor. Instead, it acts as a support tool that organizes information, detects abnormal behavior, and reduces the time needed to identify potential problems.

III. Problem Statement

The main problem addressed by Masar is the lack of intelligent real-time support for Hajj campaign supervisors. Supervisors need to monitor group members continuously and respond quickly when a pilgrim becomes separated or moves abnormally.

A basic GPS dashboard can show locations, but it does not automatically explain the risk level. Distance alone can also be misleading. A short distance may still be risky if the pilgrim is moving rapidly away, while a larger distance may be less risky if the pilgrim is returning. Therefore, the system needs temporal analysis that considers movement direction, speed, and separation trends.

The problem can be summarized into three main challenges:

- Manual tracking does not scale well in crowded environments.
- Static distance thresholds cannot fully describe movement risk.
- Supervisors need real-time alerts instead of delayed manual checking.

Masar solves this by using sequence models that process recent movement windows and detect abnormal behavior. The system is designed to answer the following question: can recent GPS-derived movement features be used to classify pilgrim movement as normal or abnormal with high reliability?

IV. Project Objectives

The main objective of Masar is to support Hajj supervisors by providing real-time tracking and automatic abnormal-movement detection. The system aims to improve awareness, reduce manual workload, and help supervisors respond faster to potential safety issues.

The project objectives are:

- Design a real-time tracking platform for supervisors and administrators.
- Collect and process GPS movement readings from pilgrims.
- Extract meaningful movement features from raw location data.
- Train and evaluate sequence models for binary anomaly detection.
- Compare Transformer, LSTM, RNN, and baseline methods.
- Integrate prediction results into a dashboard-ready workflow.
- Provide a practical foundation for future real-world Hajj deployment.

These objectives connect the technical AI component with the practical supervision problem. The model is not evaluated only as a machine learning experiment; it is evaluated as part of a system that must eventually support real operational decisions.

V. System Overview

Masar is designed as a real-time web-based tracking and supervision platform. The system includes three main user roles: administrator, supervisor, and pilgrim. Each role supports a different part of the supervision workflow.

A. Administrator

The administrator manages the system structure, including campaigns, users, and assignments. This role is important because Hajj supervision depends on organized groups and clear responsibility between supervisors and pilgrims. The administrator can create campaigns, assign pilgrims to supervisors, and control the basic data used by the system.

B. Supervisor

The supervisor monitors assigned pilgrims through a dashboard. The dashboard displays live locations, movement status, anomaly results, and alerts. The purpose is to reduce manual checking and allow the supervisor to focus on risky cases. Instead of searching through every pilgrim manually, the supervisor can prioritize pilgrims marked as abnormal by the AI module.

C. Pilgrim

The pilgrim side sends GPS data to the system. In the complete system flow, the pilgrim can also send an emergency request when help is needed. The pilgrim interface should be simple because users may have different levels of technical experience. The most important function is to provide location updates reliably.

VI. System Architecture

The Masar architecture consists of frontend, backend, database, real-time communication, and AI prediction components. These components work together to create a continuous supervision loop from GPS collection to dashboard alerting.

A. Frontend

The frontend provides dashboards for administrators and supervisors. The supervisor dashboard is the most important interface because it visualizes the current state of each pilgrim and displays alerts. It should present information in a way that is clear and fast to understand. In a crowded Hajj environment, supervisors do not have time to analyze complex data tables. Therefore, visual status labels, maps, and alert indicators are important.

B. Backend

The backend receives GPS updates, stores events, manages users and assignments, and sends movement data to the AI prediction module. It also returns prediction outputs to the dashboard. The backend acts as the connection point between the live system and the machine learning model.

C. Database

The database stores users, campaign information, assignments, GPS readings, emergency events, and anomaly predictions. A structured database design is needed so that each GPS reading can be linked to the correct pilgrim, supervisor, and campaign. This is important for both real-time monitoring and later analysis.

D. Real-Time Communication

Real-time communication is important because delayed tracking reduces the value of the system. The dashboard should update automatically when new GPS readings or anomaly predictions are available. If the system only updates after manual refresh, supervisors may miss important changes. Real-time updates allow Masar to function as a live monitoring tool rather than a static reporting platform.

E. AI Prediction Layer

The AI layer receives movement sequences and predicts whether each sequence is Normal or Abnormal. Instead of classifying one GPS point at a time, the model uses a window of recent readings. This allows the model to learn movement trends such as drifting away, sudden direction change, or unstable movement.

VII. Data Flow

The Masar data flow begins when a pilgrim device sends a GPS reading. This reading includes location and time information. The backend receives the reading and stores it in the database. After that, the system calculates derived movement features such as distance from supervisor, speed difference, and direction change.

Once enough recent readings are available, the system forms a sequence window. The window is passed to the AI prediction model. The model returns a binary prediction: Normal or Abnormal. The prediction is then saved and shown on the supervisor dashboard.

The complete flow is:

1) Pilgrim device sends GPS location.
2) Backend receives and validates the reading.
3) Reading is stored in the database.
4) Feature extraction is performed.
5) Recent movement window is prepared.
6) Transformer model predicts the movement status.
7) Dashboard updates the pilgrim status.
8) Supervisor receives visual alert if needed.

This flow keeps the system practical because the supervisor only sees the final useful result, while the technical processing happens in the background.

VIII. Dataset and Experimental Setup

The dataset was designed for binary anomaly detection on unseen routes. Each sample represents a movement sequence and is labeled as either Normal or Abnormal. The system evaluates whether the model can generalize to routes not seen during training.

The dataset contains two main labels:

- Normal: the pilgrim movement follows the expected group behavior.
- Abnormal: the pilgrim movement shows signs of separation, unusual direction, or unexpected movement behavior.

The evaluation focuses on unseen routes to test whether the model is learning movement behavior rather than memorizing specific paths. This is important because in real use, pilgrims may move through different areas, paths, and crowd conditions. A model that works only on memorized routes would not be useful for real deployment.

The experimental process includes data preprocessing, feature engineering, sequence generation, model training, model testing, and comparison against a baseline method. The same evaluation metrics were used for all models to make the comparison fair.

IX. Data Preprocessing

Before training the models, the movement data must be cleaned and prepared. GPS data can include missing values, repeated readings, irregular time intervals, and noisy points. These issues can affect model performance if they are not handled correctly.

The preprocessing steps include:

- Sorting readings by user and timestamp.
- Removing invalid or incomplete readings.
- Calculating time differences between consecutive points.
- Generating movement features from raw coordinates.
- Normalizing numerical features for model training.
- Creating fixed-length sequence windows.
- Splitting the data so that testing is performed on unseen routes.

Sorting by time is especially important because sequence models depend on the correct order of readings. If the order is wrong, the model may learn incorrect movement patterns. Normalization is also important because features such as distance and speed can have different value ranges.

X. Feature Engineering

Raw GPS coordinates are not directly enough for reliable anomaly detection. The system transforms GPS readings into movement features that better describe behavior.

The engineered features include:

- Distance between pilgrim and supervisor.
- Speed difference.
- Direction or heading difference.
- Separation rate.
- Distance trend.
- Recent movement consistency.
- Time-related changes between readings.

These features allow the model to understand how the pilgrim is moving relative to the supervisor. This is more useful than only checking the current distance.

A. Distance from Supervisor

Distance from supervisor is one of the most important features because it directly represents separation. However, it is not enough by itself. A pilgrim may be far because of temporary crowd movement but still returning toward the group.

B. Speed Difference

Speed difference helps determine whether the pilgrim is moving at a similar pace to the supervisor. If the pilgrim speed is very different, it may indicate that the pilgrim is stopping, rushing, or moving away.

C. Direction Difference

Direction difference compares the movement direction of the pilgrim with the movement direction of the supervisor. This feature helps detect cases where the pilgrim is not only separated but also moving in a different direction.

D. Separation Rate

Separation rate describes whether the distance is increasing or decreasing over time. This is important because a pilgrim who is getting farther away may require attention even before crossing a fixed distance threshold.

XI. Sequence Modeling

Movement anomaly detection is naturally a sequence problem. A single GPS reading may not show enough context, while a sequence can reveal whether the pilgrim is drifting away, returning, stopping, or moving in an unexpected direction.

The evaluated sequence models include:

- Recurrent Neural Network (RNN)
- Long Short-Term Memory (LSTM)
- Transformer

A distance-based baseline was also used for comparison. The baseline classifies a sequence as abnormal based on whether the distance exceeds a fixed threshold. This baseline is useful because it represents a simple rule-based system that could be implemented without deep learning.

XII. Baseline Method

The baseline method uses a distance-threshold rule. If the pilgrim's distance from the supervisor exceeds a predefined threshold, the movement is classified as abnormal. Otherwise, it is classified as normal.

The baseline is simple, fast, and easy to understand. However, its weakness is that it ignores the movement history. It does not know whether the pilgrim is moving back to the group, moving away from the group, or standing still. It also cannot use speed or direction patterns.

Including the baseline is important because it shows whether complex models actually provide value beyond simple distance checking. In this experiment, the baseline performed well, but the sequence models provided stronger detection behavior, especially in recall and ROC-AUC.

XIII. RNN Model

The RNN model processes the movement sequence step by step. Each time step updates the hidden state, allowing the model to carry information from previous readings. RNNs are useful for sequential data because they can represent temporal dependencies.

However, standard RNNs may struggle with longer dependencies because earlier information can become weak as the sequence grows. In this project, the RNN still performed strongly because the movement windows are not extremely long. It provides a useful comparison against LSTM and Transformer models.

XIV. LSTM Model

The LSTM model improves on standard RNNs by using gates that control what information is remembered, forgotten, and passed forward. This makes LSTM more stable for sequence learning. In movement anomaly detection, LSTM can learn patterns such as gradual separation, repeated direction changes, or sudden changes in speed.

The LSTM achieved the highest F1-score in the experiment. This shows that it balanced precision and recall very well. However, the Transformer achieved better ROC-AUC and fewer false negatives, making it especially useful for safety-focused detection.

XV. Transformer Model

The Transformer model is based on self-attention. Unlike recurrent models, the Transformer can compare all time steps in the sequence directly. This allows it to identify important changes across the movement window.

A. Input Representation

Each input sample is represented as a sequence of movement features. The sequence contains several time steps, and each time step contains engineered features such as distance, speed difference, and direction difference.

B. Self-Attention

Self-attention allows the model to decide which time steps are most important. For example, the model may focus on a sudden increase in distance, a change in direction, or a pattern showing continuous separation. This is useful because not all time steps have equal importance.

C. Classification Output

The final output is binary:

- Normal
- Abnormal

The prediction is then sent to the dashboard so that supervisors can identify risky movement.

D. Why Transformer is Suitable

The Transformer is suitable for Masar because it can focus on important moments inside the movement sequence. In pilgrim tracking, abnormal behavior may appear as a short but important change, such as a sudden turn, a rapid increase in distance, or a stop after continuous movement. Self-attention helps the model identify these important points instead of treating the sequence only as a strict step-by-step chain.

XVI. Evaluation Metrics

The models were evaluated using accuracy, precision, recall, F1-score, and ROC-AUC.

Accuracy measures the overall percentage of correct predictions. Precision measures how many predicted abnormal cases were actually abnormal. Recall measures how many actual abnormal cases were correctly detected. F1-score balances precision and recall. ROC-AUC measures the model ability to separate normal and abnormal classes across thresholds.

For safety systems, recall is especially important because missing an abnormal case can be more serious than generating a false alert. ROC-AUC is also useful because it evaluates separation quality beyond a single threshold. In Masar, a false positive may create an unnecessary alert, but a false negative may cause the supervisor to miss a risky case. Therefore, recall and false negatives are treated as important indicators.

TABLE I
Model Performance on Unseen Routes

Model	Acc.	Prec.	Recall	F1	ROC-AUC
LSTM	0.967	0.942	0.927	0.934	0.989
Transformer	0.966	0.919	0.945	0.932	0.992
RNN	0.966	0.924	0.939	0.932	0.990
Baseline	0.963	0.916	0.940	0.928	0.985

XVII. Results

Fig. 1 compares the main models using accuracy, precision, recall, F1-score, and ROC-AUC.

The results show that all sequence models performed strongly. The Transformer achieved the highest ROC-AUC score of 0.992 and the highest recall among the learned sequence models. This means it was highly effective at detecting abnormal movement cases.

The LSTM achieved the highest F1-score of 0.934, slightly above the Transformer and RNN. However, the Transformer remains important because it provides strong recall and the best separation ability according to ROC-AUC.

The results indicate that sequence-based models significantly outperform the baseline approach. While the baseline relies on a fixed distance threshold, it cannot capture temporal movement behavior such as gradual drift or sudden direction changes.

The Transformer model achieved the highest ROC-AUC score, indicating strong separation between normal and abnormal movement patterns. This suggests that the model is effective in identifying subtle behavioral changes that may not be visible through simple distance measurements.

Although the LSTM achieved a slightly higher F1-score, the Transformer produced fewer false negatives, which is critical in safety applications. Missing an abnormal case may delay intervention, making recall an important metric in this context.

XVIII. Training Behavior

Fig. 2 shows the training and validation loss curves as well as validation F1 over epochs.

The training curves show that the models improved rapidly during the early epochs and then stabilized. The validation F1 curves show that the sequence models maintained strong performance across epochs, although some fluctuation appeared due to the difficulty of abnormal movement detection.

The Transformer validation curve shows strong performance but includes a temporary drop in one epoch. This indicates that the model can be sensitive during training, but it recovers and reaches competitive performance. Such behavior is common in deep learning models, especially with time-series classification.

XIX. Confusion Matrix Analysis

Confusion matrices provide a clearer understanding of model behavior by showing true normal, false abnormal, false normal, and true abnormal predictions.

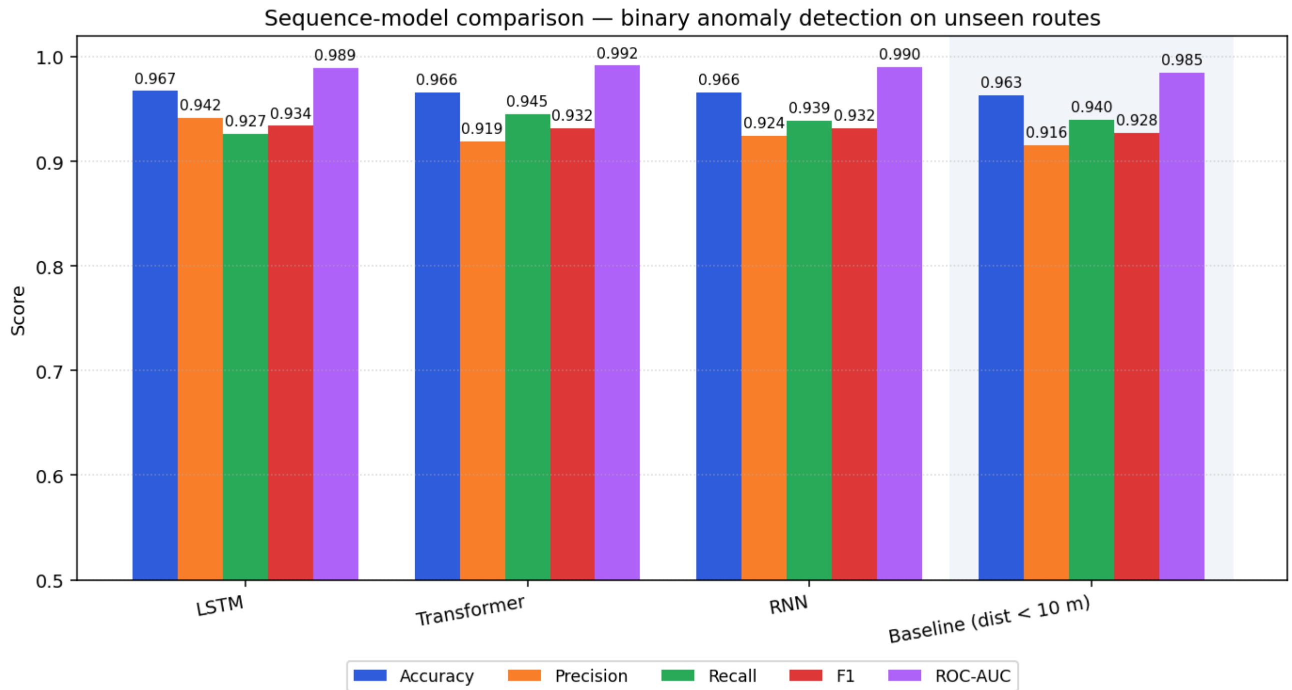

Fig. 1. Sequence-model comparison for binary anomaly detection on unseen routes.

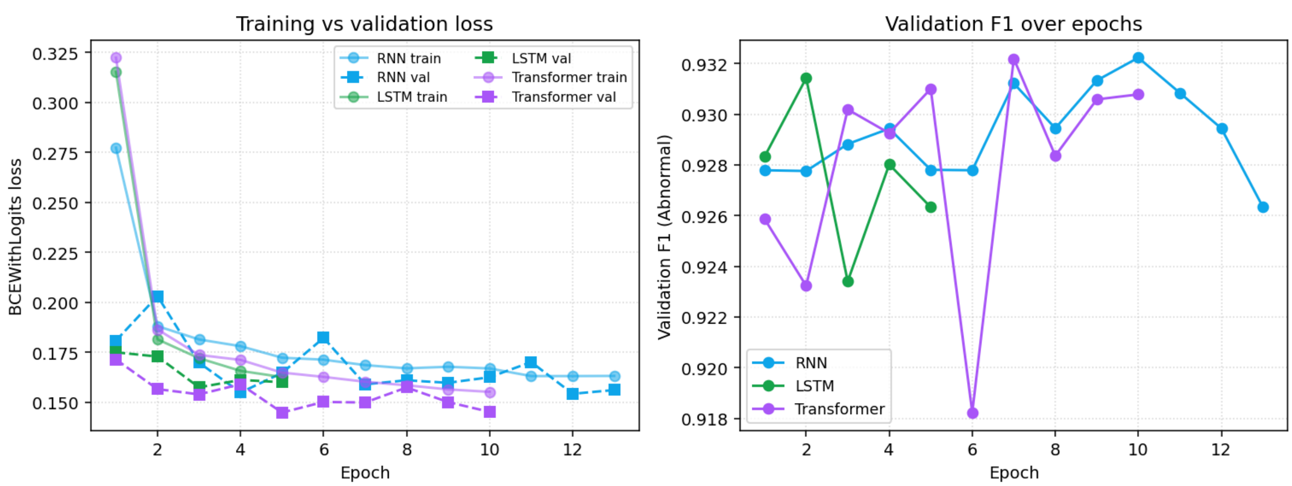

Fig. 2. Training and validation curves for the evaluated sequence models.

A. LSTM Confusion Matrix

The LSTM confusion matrix is shown in Fig. 3. The model correctly classified 10,293 normal samples and 3,220 abnormal samples. It produced 200 false positives and 255 false negatives.

B. RNN Confusion Matrix

The RNN confusion matrix is shown in Fig. 4. The model correctly classified 10,226 normal samples and 3,263 abnormal samples. It produced 267 false positives and 212 false negatives.

C. Transformer Confusion Matrix

The Transformer confusion matrix is shown in Fig. 5. The model correctly classified 10,204 normal samples and 3,285 abnormal samples. It produced 289 false positives and only 190 false negatives.

D. Baseline Confusion Matrix

The baseline confusion matrix is shown in Fig. 6. The baseline correctly classified 10,193 normal samples and 3,265 abnormal samples. It produced 300 false positives and 210 false negatives.

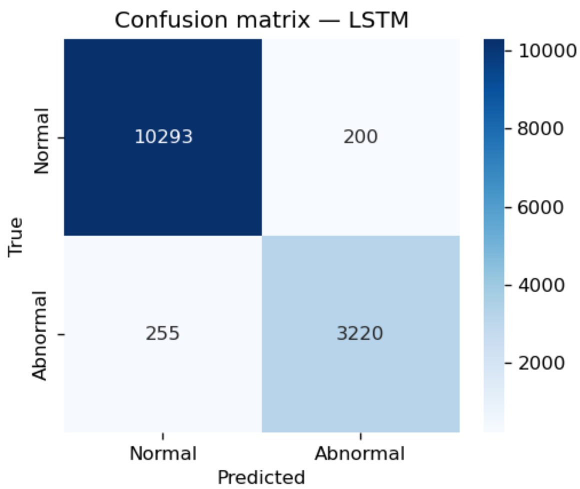

Fig. 3. Confusion matrix for the LSTM model.

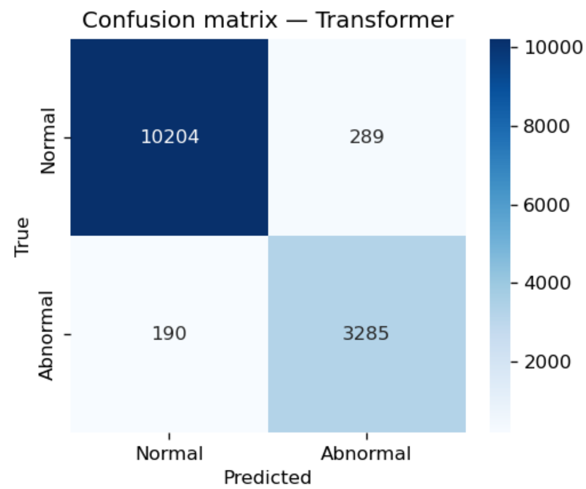

Fig. 5. Confusion matrix for the Transformer model.

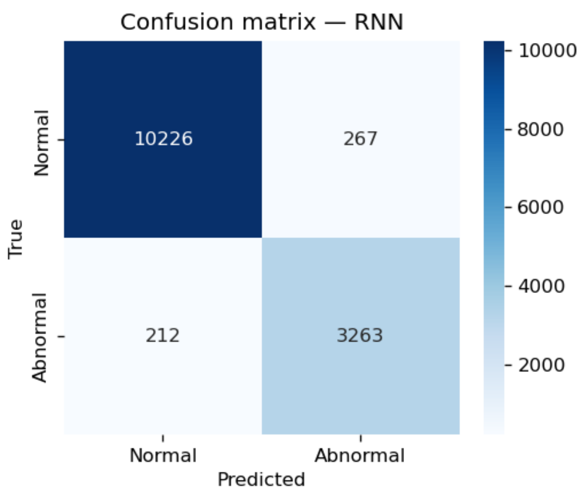

Fig. 4. Confusion matrix for the RNN model.

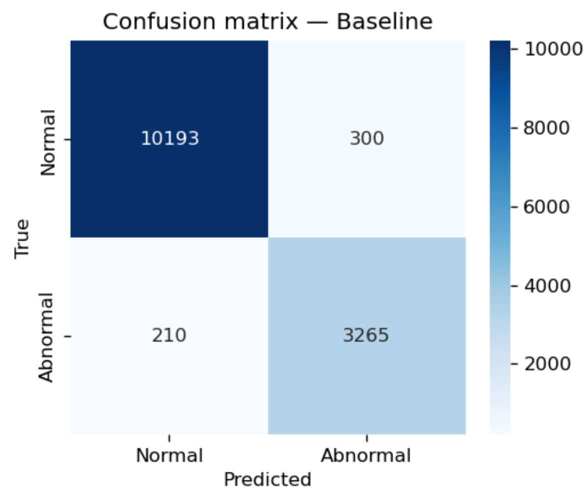

Fig. 6. Confusion matrix for the distance-threshold baseline.

The Transformer produced the lowest number of false negatives among the evaluated methods. This is important because false negatives represent abnormal cases that the system failed to detect. In a safety-focused system, reducing false negatives is critical because missed abnormal movement may delay supervisor response.

XX. Real-Time System Integration

The prediction model can be integrated into the real-time Masar system using a sliding-window approach. Each pilgrim has a recent sequence of movement readings. When a new GPS reading arrives, the system updates the sequence and sends it to the model for prediction.

The supervisor dashboard receives the output and displays the current status of each pilgrim. If abnormal movement is detected, the system can highlight the pilgrim or generate an alert.

The real-time flow is:

- Pilgrim sends GPS reading.
- Backend stores the reading.
- Feature extraction is applied.
- Sequence window is updated.
- Transformer predicts Normal or Abnormal.
- Dashboard updates the pilgrim status.

This integration makes the AI model operational. The model is not used only after the trip ends; it can support live decision-making. The sliding-window method is also practical because it does not require the system to process the entire movement history every time.

XXI. Dashboard and Alerts

The dashboard is designed to help supervisors understand the status of pilgrims quickly. Instead of only displaying raw coordinates, the dashboard should display useful status information.

The dashboard can include:

- Live pilgrim locations.
- Normal and abnormal status labels.
- Alert notifications.
- SOS requests.
- Search and filtering tools.
- Assignment information.

This helps the supervisor focus on pilgrims who require attention rather than manually checking every location point. The dashboard should also avoid visual overload. If too many alerts appear without organization, the supervisor may ignore them. Therefore, alerts should be ranked or filtered based on severity, time, and assigned group.

XXII. Alert Prioritization

In real-world supervision, not all alerts have the same level of urgency. A pilgrim who is slightly out of range but returning to the group is less urgent than a pilgrim who is moving away quickly or has stopped for a long time. Therefore, Masar can support alert prioritization.

Possible alert levels include:

- Low: unusual movement is detected, but the pilgrim remains close.
- Medium: separation is increasing or movement direction differs from the group.
- High: the pilgrim is far, moving away, inactive, or sending an SOS request.

This prioritization can help supervisors respond in the correct order. It also reduces alert fatigue by preventing every abnormal prediction from being treated as equally critical.

XXIII. Deployment Considerations

Deploying the Masar system in a real-world Hajj environment requires careful consideration of scalability, reliability, and data privacy. The system must handle large numbers of simultaneous users, including supervisors and pilgrims, while maintaining low latency for real-time updates.

One key requirement is backend scalability. A production deployment would require a distributed backend architecture with load balancing and a high-performance database system such as PostgreSQL instead of SQLite. This ensures that the system can handle thousands of concurrent location updates without performance degradation.

Another important aspect is network reliability. Pilgrim devices may experience unstable connections due to crowd density or infrastructure limitations. Therefore, the system should include buffering mechanisms and fallback strategies to prevent data loss. For example, the device can temporarily store GPS readings and send them when the connection becomes available again.

Privacy and security are also critical. Since the system processes location data, it must ensure secure communication, authentication, and controlled access to sensitive information. Data encryption and role-based permissions are essential to protect user privacy.

Finally, user experience must be considered. The dashboard should remain simple and responsive, allowing supervisors to quickly identify abnormal cases without being overwhelmed by excessive information.

XXIV. Security and Privacy

Location data is sensitive because it shows where a person is and how they move. For this reason, Masar must be designed with privacy protection from the beginning. Only authorized supervisors should be able to view pilgrims assigned to them. Administrators should manage permissions carefully, and system logs should be protected.

Security measures may include:

- User authentication for administrators and supervisors.
- Role-based access control.
- Encrypted communication between device, backend, and dashboard.
- Secure storage of sensitive records.
- Limited access to historical location data.
- Audit logs for important actions.

These measures are important because a tracking system can create risk if data is exposed or misused. A safe system should not only detect abnormal movement but also protect the privacy of the people being monitored.

XXV. Discussion

The results show that sequence-based models are effective for binary anomaly detection in pilgrim movement. The Transformer performed especially well in recall and ROC-AUC, meaning it was strong at detecting abnormal cases and separating the two classes.

The LSTM achieved the highest F1-score, but the Transformer achieved fewer false negatives. In the context of Hajj supervision, false negatives are particularly important because they represent risky cases that the system missed. Therefore, the Transformer is a strong candidate for safety-focused deployment.

The baseline also performed well, which shows that distance is an important signal. However, the sequence models provide a more flexible approach because they can analyze movement behavior over time. A threshold can detect obvious separation, but it cannot fully understand whether the pilgrim is drifting, returning, stopping, or changing direction.

The experiment also shows that model selection should depend on the goal of the system. If the goal is balanced performance, LSTM is very strong. If the goal is safety and reducing missed abnormal cases, the Transformer is more suitable because of its low false negatives and high ROC-AUC.

XXVI. Practical Impact

Masar can improve the practical workflow of Hajj campaign supervision. Instead of waiting for manual reports or repeated headcounts, supervisors can receive live status updates. This can reduce confusion, improve response time, and help supervisors manage larger groups with more confidence.

The system can also create historical movement records that help campaigns evaluate their operations after each trip. For example, supervisors can review which routes caused more abnormal movement, which areas had delayed updates, and which times were most difficult to manage. This can support better planning for future campaigns.

Masar is also useful beyond Hajj. Similar tracking and anomaly detection concepts can be applied to Umrah groups, school trips, large events, sports events, and any situation where group movement must be supervised safely.

XXVII. Limitations

The first limitation is that the dataset is simulated or controlled rather than fully collected from real Hajj conditions. Real GPS signals may include more noise, missing readings, and device differences.

The second limitation is that binary labels simplify the problem into Normal and Abnormal. In a future system, it may be useful to separate abnormal cases into warning, deviation, delay, and critical states.

The third limitation is the need for stable GPS updates. If updates are delayed or inaccurate, feature extraction and sequence prediction may become less reliable.

The fourth limitation is scalability. A full deployment would require load testing with many simultaneous pilgrims and supervisors.

The fifth limitation is that abnormal movement can have different meanings depending on the situation. A pilgrim may be separated because of crowd flow, illness, transportation, or a planned route change. The model can detect unusual behavior, but human supervisors still need to make the final decision.

XXVIII. Future Work

Future work should focus on improving real-world readiness. The most important step is collecting real GPS movement data under controlled and ethical conditions. This would allow the model to learn from real noise and natural movement behavior.

Future improvements include:

- Testing on real Hajj or Hajj-like movement data.
- Expanding binary detection into multi-class risk levels.
- Improving GPS noise filtering.
- Adding route history and heatmap visualization.
- Enhancing dashboard search and filtering.
- Performing large-scale real-time backend testing.
- Adding supervisor feedback to improve future predictions.
- Supporting offline buffering when network connection is weak.
- Creating multilingual interfaces for different pilgrim groups.

A future version of Masar can also include model explainability. For example, the system could explain that an abnormal alert happened because distance increased quickly, direction changed, or the pilgrim stopped moving. This would help supervisors trust the model and understand the reason behind each alert.

XXIX. Conclusion

This paper presented Masar, a real-time pilgrim tracking and safety system using sequence-based anomaly detection. The system is designed to support Hajj supervisors by detecting abnormal pilgrim movement based on GPS-derived features.

The Transformer model achieved strong results, including 0.966 accuracy, 0.945 recall, 0.932 F1-score, and the highest ROC-AUC of 0.992. It also produced the fewest false negatives among the evaluated methods. These results show that Transformer-based sequence modeling is a promising approach for safety-focused pilgrim supervision.

Masar provides a strong foundation for intelligent crowd monitoring during Hajj and similar large-scale events. By combining real-time GPS tracking, engineered movement features, AI prediction, and dashboard alerts, the system can help supervisors identify risky situations earlier and manage group movement more effectively.

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